

Jack Duvall - Final Project - Section 205

Boolean Expressions: F = Front Door

B = Back Door

W = Window

A = ARMED

Ex. 1 ~ All Entry Point Closed.

$$F = 0 \quad A = \neg(0 + 0 + 0)$$

$$B = 0 \quad A = \neg(0)$$

$$W = 0 \quad \boxed{A = 1}$$

Ex. 2 ~ Front door Open.

$$F = 1 \quad A = \neg(1 + 0 + 0)$$

$$B = 0 \quad A = \neg(1)$$

$$W = 0 \quad \boxed{A = 0}$$

Ex. 3 ~ Window Open.

$$F = 0 \quad A = \neg(0 + 0 + 1)$$

$$B = 0 \quad A = \neg(1)$$

$$W = 1 \quad \boxed{A = 0}$$

Ex. 4 ~ Back door Open.

$$F = 0 \quad A = \neg(0 + 1 + 0)$$

$$B = 1 \quad A = \neg(1)$$

$$W = 0 \quad \boxed{A = 0}$$

Ex. 5 ~ All Entry Points Open.

$$F = 1 \quad A = \neg(1 + 1 + 1)$$

$$B = 1 \quad A = \neg(1)$$

$$W = 1 \quad \boxed{A = 0}$$